

Signature Methods for Optimal Market Making

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Abstract

We propose a signature-based method to solve the optimal market-making problem under a mean-variance criterion. By exploiting signature linearization techniques, we reduce the market-making problem to a pseudo-linear optimization over the expected signature of an augmented market path, and we develop a signature algorithm named Sig-REINFORCE to learn the optimal bid and ask quotes. We test our method in two scenarios, in which market-order arrivals follow either a Poisson or a self-exciting Hawkes process, and we benchmark it against a Proximal Policy Optimization (PPO) baseline.

1 Introduction

1.1 The market making problem

The seminal work of Ho and Stoll [Ho and Stoll \(1981\)](#) set the foundations of the market making problem, in which a trader optimally sets their bid/ask quotes on a limit order book, facing a continuous-time order flow modeled with Poisson jump processes and facing return risk on their stock and on the rest of their portfolio modeled by diffusion processes. It was then mathematically formalized in [Avellaneda and Stoikov \(2008\)](#); [Stoikov and Sauglam \(2009\)](#); [Guéant et al. \(2012\)](#); [Euch et al. \(2021\)](#), in which the price of a risky asset is modeled through a Bachelier (Brownian) model with Poisson order flow. It was then extended to option pricing in the Heston model [Baldacci et al. \(2021a\)](#), the rough Heston model [Rosenbaum and Zhang \(2022\)](#), multiple marker makers [Baldacci et al. \(2021b\)](#), persistent or temporary noise components [Barucci et al. \(2025\)](#) with Poisson order flow, and to Hawkes processes [Jusselin \(2021\)](#). We also refer to the monographs [O’hara \(1998\)](#); [Cartea et al. \(2015\)](#) for more details. Most of these studies cannot, however, reflect the complexity of both order flows and the fundamental price, nor cover all the empirical facts observed in financial markets, quoting Rama Cont in ([Cont, 2001](#), Conclusion): “The properties mentioned here [in [Cont \(2001\)](#)] are model free in the sense that they do not result from a parametric hypothesis on the return process but from rather general hypotheses of qualitative nature. As such, they should be viewed as constraints that a stochastic process has to verify in order to reproduce the statistical properties of returns accurately. Unfortunately, most currently existing models fail to reproduce all these statistical features at once, showing that they are indeed very constraining”. Model-free or “let the data speak for themselves” methods for market making using reinforcement learning techniques have been well developed in the last decades, from the foundations in [Nevmyvaka et al. \(2006\)](#),

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extended to the mean-variance criterion, see for example [Wang and Zhou \(2020\)](#), multi-agent settings [Wang et al. \(2025\)](#), adversarial reinforcement learning [Zelman et al. \(2024\)](#), or more recently anticipating closing auctions in adaptive market making [Graf and Mastrolia \(2026\)](#). We also refer to [Beysolow II \(2019\)](#); [Hambly et al. \(2023\)](#) for reinforcement learning techniques in market making and machine learning techniques used in finance.

On the one hand, the model-based framework provides a rigorous framework and theoretical existence or convergence results for the problem, easily tractable for numerical simulations, while the convergence of RL algorithms remains difficult in practice, with ‘post-convergence instability’ challenges among others, see [Dulac-Arnold et al. \(2021\)](#). On the other hand, model-based methods are inherently restrictive: the perfect model does not exist, while RL methods literally let the data speak for themselves and so incorporate by nature the specificity of any market. The goal of this work is to find a middle ground to keep the best of both worlds: a mathematically rigorous method while keeping the properties of the process rather than setting a mathematical model in the optimization problem. A natural tool for this purpose is the signature of a stochastic process, which has already been successfully used in mathematical finance.

1.2 Signature methods: background and motivation

This work proposes a signature-based approximation framework for path-dependent market making. The signature of a path - the collection of all its iterated integrals - originates in the work of [Chen \(1957\)](#) and plays a fundamental role in rough path theory (see [Lyons \(1998\)](#); [Friz and Hairer \(2014\)](#)). More recently, signatures have emerged as a powerful tool in machine learning and mathematical finance for extracting informative features from time-series data. Its application areas can roughly be grouped into the following ones: optimal control (see [Bayer et al. \(2023\)](#); [Bank et al. \(2025\)](#)), data generation/simulation (see [Bühler et al. \(2020\)](#); [Ni et al. \(2022\)](#); [Liao et al. \(2024\)](#)), financial modeling (see [Abi Jaber and Gérard \(2024\)](#); [Abi Jaber et al. \(2024, 2025a,b\)](#); [Andrès et al. \(2024\)](#); [Arribas et al. \(2021\)](#); [Benth et al. \(2024\)](#); [Cartea et al. \(2022\)](#); [Cuchiero et al. \(2023, 2024\)](#); [Kalsi et al. \(2020\)](#); [Lyons et al. \(2020\)](#)), signature kernel regression methods and applications to path-dependent PDEs in finance (see [Salvi et al. \(2021\)](#); [Lee and Oberhauser \(2023\)](#); [Pannier and Salvi \(2024\)](#)). Additionally, signatures have successfully been employed at Amazon to forecast transportation marketplace rates [Gu et al. \(2024\)](#), and to electricity demand forecasting [Drobac et al. \(2025\)](#).

Recent new developments of notion of signature have also emerged (see [Abi Jaber and Sotnikov \(2026\)](#); [Bloch et al. \(2026\)](#); [Hager et al. \(2026\)](#)).

From an approximation-theoretic perspective, iterated integrals play the role of monomials on path space. They provide a natural basis for representing path-dependent functionals, and many of the familiar properties of polynomial approximation extend to signatures. In particular:

- *Algebra.* Products of signature-linear functionals are again signature-linear, via the shuffle product (see equation (2.1)).
- *Universal approximation.* Linear functionals of the signature are dense in suitable spaces of continuous path functionals (see e.g., [Kalsi et al. \(2020\)](#); [Bayer et al. \(2023\)](#); [Benth et al. \(2024\)](#); [Cuchiero et al. \(2025b\)](#)).
- *Taylor expansions.* Sufficiently regular non-anticipative path functionals admit expansions in terms of signature coordinates (see [Cuchiero et al. \(2025a\)](#); [Dupire and Tissot-Daguet \(2023\)](#); [Bandi et al. \(2025\)](#)).

These properties make signature-linear functionals both expressive and tractable as policy classes for stochastic control. Since only a linear readout must be optimized, a broad class of non-Markovian control problems can be reduced to finite-dimensional optimization. Signature methods have consequently been applied successfully to a range of problems in mathematical finance, including portfolio optimization and algorithmic trading (see e.g., [Akyildirim et al. \(2025\)](#); [Cartea et al. \(2022\)](#); [Cuchiero and Möller \(2025\)](#); [Futter et al. \(2023\)](#); [Kalsi et al. \(2020\)](#); [Lyons et al. \(2020\)](#)).

While signature methods have become an established tool for continuous-path systems, their extension to càdlàg paths and semimartingales with jumps is comparatively recent, and practical applications remain largely unexplored. This work develops a signature-based approximation framework for path-dependent market making and, to the best of our knowledge, provides the first applied use of signatures for stochastic control problems involving jump processes. The market-making setting considered here offers a natural and important testbed for this emerging theory, since order-flow dynamics are inherently discontinuous and are typically modeled by counting processes.

The signature of a càdlàg rough path was introduced in [Friz and Victoir \(2017\)](#), where the classical iterated-integral construction is modified to account for the interaction between continuous and jump components. The corresponding universal approximation theorem for càdlàg paths was subsequently established in [Cuchiero et al. \(2025b\)](#). This result is essential for the present work, as the order-flow processes driving market-making dynamics are inherently discontinuous.

Specifically, the market maker posts ask and bid offsets δ^a and δ^b in response to buy and sell market orders represented by counting processes N^a and N^b .

Our approach is based on a signature representation of path-dependent controls. Let $\mathbb{Y}_t$ denote the signature of an augmented market path up to time t , incorporating time, price information, quadratic-variation covariates, and order-flow history (see [Section 2.4](#) for a precise definition). We parameterize the ask and bid controls as

$$\delta_t^a = \langle \alpha^a, \mathbb{Y}_t \rangle, \quad \delta_t^b = \langle \alpha^b, \mathbb{Y}_t \rangle,$$

where α^a and α^b are trainable tensors supported on words up to a prescribed truncation level. This choice is not merely a computational convenience. By the càdlàg approximation theorem of [Cuchiero et al. \(2025b\)](#), signature-linear functionals form a universal class for sufficiently regular non-anticipative controls, implying that any such control can be approximated arbitrarily well within this parameterization.

A second advantage of the signature framework is that it renders the optimization objective highly tractable. Using the algebraic structure of signatures, stochastic integrals of linear functionals of the signature against counting processes can be represented, after an explicit correction via the *tilde transform* ([Definition 2.4](#)), as linear functionals of the terminal signature. Nonlinear interactions are handled through the shuffle product identity, which expresses products of signature features as elements of the same algebra. As a result, the realized trading payoff admits a finite-dimensional polynomial representation in the control parameters.

Following this reformulation, the principal source of complexity is no longer the payoff functional itself, but the dependence of the controlled probability law on the policy through the order-arrival intensities. We address this challenge using a likelihood-ratio gradient estimator, which enables direct optimization over a rich class of path-dependent controls while preserving computational efficiency.

1.3 Contributions and outline

The contributions of this work are fourfold and listed below.

- We formulate the first signature-linear class of path-dependent market-making controls for semimartingale price and point-process order-flow dynamics. As far as we know, this is also the first application of the signature method to optimal control with jump processes.
- We provide a comprehensive and explicit transformation which rewrites the relevant stochastic integrals driven by continuous and counting processes (modeling the price and order-arrival components respectively) as linear functionals of the signature of an augmented path.
- We show how the mean–variance objective investigated can be reduced, via the shuffle product, to an algebraic functional of the expected terminal signature under the controlled law.
- We combine this representation with a likelihood-ratio gradient for spread-dependent point-process intensities and score functions.

The outcome of this study is to provide a fully tractable Sig-REINFORCE algorithm taking as input the signature of the observed data and returning as output the optimized bid/ask quotes that maximize the mean-variance market-making objective. The structure of this study is the following. In Section 2, we recall the main notation and results from signature theory and we present the general problem and the main linearization results obtained from the signature method. Section 3 applies these results to the mean-variance problem investigated, using shuffle reduction to linearize the problem and the likelihood-ratio gradient method to explore the objective function and reach the critical points (optimal bid/ask quotes), and provides the Sig-REINFORCE algorithm. Finally, Section 4 shows numerical results in Poisson and Hawkes order-flow environments, and we compare our Sig-REINFORCE method with the popular on-policy PPO, see [Schulman et al. \(2017\)](#). In the Poisson model, where the expert benchmark is tailored to the memoryless structure, the signature policy recovers the qualitative behavior of the benchmark with a modest loss in mean reward and outperforms PPO. In the Hawkes model, where order flow is self-exciting and the control problem is more genuinely path-dependent, the value generated by the signature policy stays close to that of PPO while reducing the computational effort.

2 Signature market making optimization

2.1 Notations

Throughout this study we consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that Ω is the product of the Wiener space and the Skorokhod space $\mathcal{C}([0, T]; \mathbb{R}) \times \mathbb{D}([0, T]; \mathbb{N}) \times \mathbb{D}([0, T]; \mathbb{N})$ where $\mathcal{C}([0, T]; \mathbb{R})$ and $\mathbb{D}([0, T]; \mathbb{R})$ denote the space of continuous and càdlàg functions from $[0, T]$ into $\mathbb{R}$ or $\mathbb{N}$ respectively.

We now collect the notations and background on signatures used in this work. Fix $d \in \mathbb{N}$ and let $\mathbb{R}^d$ be the Euclidean space. The extended tensor algebra over $\mathbb{R}^d$ is defined by

$$T((\mathbb{R}^d)) := \left\{ \mathbf{u} = (\mathbf{u}^{(0)}, \mathbf{u}^{(1)}, \dots, \mathbf{u}^{(n)}, \dots) \mid \mathbf{u}^{(n)} \in (\mathbb{R}^d)^{\otimes n} \right\},$$

where $(\mathbb{R}^d)^{\otimes n}$ denotes the n -fold tensor product of $\mathbb{R}^d$, with the convention $(\mathbb{R}^d)^{\otimes 0} := \mathbb{R}$. We equip $T((\mathbb{R}^d))$ with the standard addition $+$, tensor multiplication $\otimes$, and scalar multiplication. For $N \in \mathbb{N}$, the truncated tensor algebra is defined by

$$T^N(\mathbb{R}^d) := \left\{ \mathbf{u} = (\mathbf{u}^{(0)}, \mathbf{u}^{(1)}, \dots, \mathbf{u}^{(N)}) \mid \mathbf{u}^{(n)} \in (\mathbb{R}^d)^{\otimes n} \text{ for } n \leq N \right\}.$$

Let $\pi_n : T((\mathbb{R}^d)) \rightarrow (\mathbb{R}^d)^{\otimes n}$ be the map such that, for $\mathbf{u} \in T((\mathbb{R}^d))$, $\pi_n(\mathbf{u}) = \mathbf{u}^{(n)}$, and let $\pi_{\leq N} : T((\mathbb{R}^d)) \rightarrow T^N(\mathbb{R}^d)$ be such that, for $\mathbf{u} \in T((\mathbb{R}^d))$,

$$\pi_{\leq N}(\mathbf{u}) = \mathbf{u}^N := (\mathbf{u}^{(n)})_{n=0}^N.$$

Let $I = (i_1, \dots, i_n)$ be a multi-index with entries in $\{1, \dots, d\}$. Denoting by $\epsilon_1, \dots, \epsilon_d$ the canonical basis of $\mathbb{R}^d$, we use the notation $|J| := n$ and

$$\epsilon_J := \epsilon_{j_1} \otimes \epsilon_{j_2} \otimes \dots \otimes \epsilon_{j_n}.$$

Observe that $(\epsilon_J)_J$ is the canonical orthonormal basis of $(\mathbb{R}^d)^{\otimes n}$. Furthermore, we denote by $\epsilon_{\emptyset}$ the basis element of $(\mathbb{R}^d)^{\otimes 0}$ and set $|\emptyset| := 0$. We also set $J' := (j_1, \dots, j_{n-1})$ for $n > 1$, $J' := \emptyset$ for $n = 1$, and $J'' := (J')'$ for $n > 1$, with the convention $\epsilon_{J''} = 0$ for $n = 1$.

For two multi-indices $J \in \{1, \dots, d\}^{|J|}$ and $L \in \{1, \dots, d\}^{|L|}$, and two indices of length 1, $a, b \in \{1, \dots, d\}$, the shuffle product $\sqcup$ is defined recursively by

$$\begin{aligned} J \sqcup \emptyset &= \emptyset \sqcup J = J, \\ (J, a) \sqcup (L, b) &= ((J \sqcup (L, b)), a) + (((J, a) \sqcup L), b), \end{aligned} \tag{2.1}$$

where (J, a) denotes the concatenation of multi-indices. Given $\mathbf{x} \in T((\mathbb{R}^d))$, we write $\mathbf{x}_J := \langle \mathbf{x}, \epsilon_J \rangle$, and for each $\mathbf{u} \in T(\mathbb{R}^d)$, we set

$$\langle \mathbf{u}, \mathbf{x} \rangle := \sum_{|J| \geq 0} \mathbf{u}_J \mathbf{x}_J \in \mathbb{R}. \tag{2.2}$$

Notice that the sum in (2.2) is finite by definition of $T(\mathbb{R}^d)$. Define

$$G((\mathbb{R}^d)) := \left\{ \mathbf{x} \in T((\mathbb{R}^d)) : \langle \epsilon_{\emptyset}, \mathbf{x} \rangle = 1, \langle \epsilon_J \sqcup \epsilon_L, \mathbf{x} \rangle = \langle \epsilon_J, \mathbf{x} \rangle \langle \epsilon_L, \mathbf{x} \rangle, \text{ for all } J, L \right\}. \tag{2.3}$$

2.2 Signature of càdlàg semimartingales

In this section, we recall the notion of the signature of càdlàg semimartingale, as introduced in [Friz and Atul \(2017\)](#). More detailed discussion can be found in [\(Primavera, 2024, Chapter 1\)](#).

Definition 2.1. Fix $T > 0$. For a semimartingale $X = (X_t)_{t \in [0, T]}$, we call the $G((\mathbb{R}^d))$ -valued process $\mathbb{X} = (\mathbb{X}_t)_{t \in [0, T]}$ the *signature of X* if $\langle \epsilon_{\emptyset}, \mathbb{X}_t \rangle = 1$ and

$$\begin{aligned} \langle \epsilon_J, \mathbb{X}_t \rangle &= \int_0^t \langle \epsilon_{J'}, \mathbb{X}_{s-} \rangle dX_s^{i_{|J|}} 1_{\{|J| > 0\}} + \frac{1}{2} \int_0^t \langle \epsilon_{J''}, \mathbb{X}_{s-} \rangle d[X^{i_{|J|-1}}, X^{i_{|J|}}]_s^c 1_{\{|J| > 1\}} \\ &\quad + \sum_{0 < s \leq t} \sum_{\epsilon_{J_1} \otimes \epsilon_{J_2} = \epsilon_J} \frac{1}{|J_2|!} 1_{\{|J_2| > 1\}} \langle \epsilon_{J_1}, \mathbb{X}_{s-} \rangle \langle \epsilon_{J_2}, (\Delta X_s)^{\otimes |J_2|} \rangle, \end{aligned}$$

for each $|J| > 0$.

Remark 2.2. When X is a continuous semimartingale, the signature introduced above reduces to the Stratonovich signature studied, for instance, in [Cuchiero et al. \(2023\)](#); [Abi Jaber et al. \(2025a\)](#).

2.3 Market making problem

We consider a market maker trading a financial instrument whose mid-price process is denoted by S , over an intraday horizon $[0, T]$; interest rates are neglected throughout. At each time t , the market maker posts an ask price S_t^a and a bid price S_t^b , or equivalently controls the half-spreads

$$\delta_t^a := S_t^a - S_t, \quad \delta_t^b := S_t - S_t^b,$$

which we refer to as the ask and bid offsets, respectively.

Let N^a and N^b be counting processes representing the cumulative arrivals of market orders hitting the ask and the bid, respectively, with intensities λ^a and λ^b . We assume that these intensities depend on the corresponding offsets:

$$\lambda_t^a = h^a(\delta_{t-}^a), \quad \lambda_t^b = h^b(\delta_{t-}^b), \quad (2.4)$$

for some suitable positive valued functions h^a, h^b . The inventory process is defined as

$$Q_t := N_t^b - N_t^a,$$

and tracks the net position accumulated by the market maker up to time t . The profit and loss generated over the horizon $[0, T]$ decomposes as

$$\int_0^T (S_t^a dN_t^a - S_t^b dN_t^b) + (Q_T S_T - Q_0 S_0) = Z_T^\delta + I_T,$$

where, for $\delta := (\delta^a, \delta^b)$, Z^δ is the execution revenue, capturing the spread income earned on each trade, and I is the inventory P&L, driven by fluctuations in the mid-price:

$$dZ_t^\delta := \delta_t^a dN_t^a + \delta_t^b dN_t^b, \quad dI_t := Q_t dS_t.$$

The market maker's problem is to choose the posting strategy δ so as to maximize the expected utility of the total P&L:

$$V_0^{MM} = \sup_{\delta = (\delta^a, \delta^b)} \mathbb{E} \left[U \left(Z_T^\delta, I_T, (I_T)^2 \right) \right], \quad (2.5)$$

for some utility function U to be specified.

Remark 2.3. *At this stage we deliberately leave unspecified both the dynamics of the mid-price S and the order-arrival processes N^a, N^b , as well as the precise form of the utility function U . These will be made concrete in subsequent sections.*

2.4 Signature methods

We show that, by restricting to the class of signature-linear offsets introduced below, the optimization problem (2.5) admits a tractable reformulation in terms of the signature of a state process, which summarizes the relevant market information up to the current time.

State process and notation. Without loss of generality, we assume that $\Delta N_t^a \Delta N_t^b = 0$ for all t and define the state process

$$Y_t := \left(t, S_t, [S]_t, N_t^a, \sum_{s \leq t} (\Delta N_s^a)^2, \dots, \sum_{s \leq t} (\Delta N_s^a)^M, N_t^b, \sum_{s \leq t} (\Delta N_s^b)^2, \dots, \sum_{s \leq t} (\Delta N_s^b)^M \right),$$

for some $M \geq 2$, whose components we denote by

$$Y_t^0 := t, \quad Y_t^s := S_t, \quad Y_t^{qs} := [S]_t, \quad Y_t^a := N_t^a, \quad Y_t^b := N_t^b,$$

and

$$Y_t^{a^m} := \sum_{s \leq t} (\Delta N_s^a)^m, \quad Y_t^{b^m} := \sum_{s \leq t} (\Delta N_s^b)^m,$$

for $2 \leq m \leq M$. Multi-indices have entries in the alphabet

$$\mathcal{A} := \{0, s, qs, a, a^2, \dots, a^M, b, b^2, \dots, b^M\},$$

with $|\mathcal{A}| = 3 + 2M$, and the notation of Section 2.1 applies throughout.

Linearization of stochastic integrals. To linearize stochastic integrals via the signature, we introduce the following transformation on tensors.

Definition 2.4. For each multi-index J with entries in $\{0, s, a, b\}$ and each $j \in \{0, s, a, b\}$, define

$$\begin{aligned} (\epsilon_J; \epsilon_j)^\sim &:= \epsilon_J \otimes \epsilon_j - \frac{1}{2} \epsilon_{J'} \otimes \epsilon_{qs} \mathbf{1}_{\{|J| = j = s\}} \\ &+ \sum_{\epsilon_{J_1} \otimes \epsilon_{J_2} = \epsilon_J} \alpha(|J_2|) (\epsilon_{J_1} \otimes \epsilon_{a|J_2|+1}) \mathbf{1}_{\{j=a\}} \mathbf{1}_{\{J_2=(a, \dots, a)\}} \\ &+ \sum_{\epsilon_{J_1} \otimes \epsilon_{J_2} = \epsilon_J} \alpha(|J_2|) (\epsilon_{J_1} \otimes \epsilon_{b|J_2|+1}) \mathbf{1}_{\{j=b\}} \mathbf{1}_{\{J_2=(b, \dots, b)\}}, \end{aligned}$$

where $\alpha(r) := \sum_{k=1}^r (-1)^k \alpha(r, k)$ and $\alpha(r, k) := \sum_{J \in \mathcal{I}(r, k)} \prod_{i=1}^k \frac{1}{(j_i+1)!}$, with $\mathcal{I}(r, k)$ denoting the set of multi-indices $J \in \mathbb{N}^k$, with $j_i \geq 1$ for every i , summing to r .

Remark 2.5. Under the additional assumption $[S]_t = \sigma^2 t$, the term $\epsilon_{J'} \otimes \epsilon_{qs}$ collapses to $\sigma^2 \epsilon_{J'} \otimes \epsilon_0$. If furthermore all jumps are of unit size, $\Delta N_t^a = \Delta N_t^b = 1$ at every jump time, then $\epsilon_{a|J_2|+1}$ and $\epsilon_{b|J_2|+1}$ reduce to ϵ_a and ϵ_b , yielding a considerably simpler expression. Both simplifications are exploited in the subsequent sections. We also refer to Section 4.2 of [Cuchiero et al. \(2025b\)](#) for the extension of Definition 2.4 to general index sets.

The following proposition is the cornerstone of our approach: it shows that Itô integrals of signature-linear functionals remain signature-linear. The proof is a straightforward adaptation of Proposition 4.10 in [Cuchiero et al. \(2025b\)](#).

Proposition 2.6. Fix $n \in \mathbb{N}$, $J \in \{0, s, a, b\}^n$, and $j \in \{0, s, a, b\}$. If $M \geq n + 1$, then

$$\int_0^t \langle \epsilon_J, \mathbb{Y}_{s-} \rangle dY_s^j = \langle (\epsilon_J; \epsilon_j)^\sim, \mathbb{Y}_t \rangle.$$

By linearity of the stochastic integral, this extends immediately to $\mathbf{u} \in T(\mathbb{R}^4)$: for every $j \in \{0, s, a, b\}$,

$$\int_0^t \langle \mathbf{u}, \mathbb{Y}_s \rangle dY_s^j = \langle (\mathbf{u}; \epsilon_j)^\sim, \mathbb{Y}_t \rangle, \quad (\mathbf{u}; \epsilon_j)^\sim := \sum_{|J| \geq 0} u_J (\epsilon_J; \epsilon_j)^\sim.$$

In other words, Itô integrals of linear functionals of $\mathbb{Y}$ against components of Y are again linear functionals of $\mathbb{Y}$.

Signature-linear offsets and problem reformulation. We now introduce the class of signature-linear offsets, i.e., controls parametrised as linear functionals of the signature of Y , and show how Proposition 2.6 yields a tractable reformulation of (2.5). The restriction to this class is natural and well-motivated: by the universal approximation properties of the signature of càdlàg paths (see Cuchiero et al. (2025b)), any sufficiently regular control can be approximated arbitrarily well by a signature-linear one.

Definition 2.7 (Lin-Sig-Offsets). Let $K \in \mathbb{N}$ be fixed. The space of *signature-linear offsets* is $\mathcal{D}^{\text{sig}} := \mathcal{D}^{\text{sig},a} \times \mathcal{D}^{\text{sig},b}$, where

$$\begin{aligned}\mathcal{D}^{\text{sig},a} &:= \left\{ \delta_t^a = \langle \alpha^a, \mathbb{Y}_t \rangle := \sum_{|J| \leq K} \alpha_J^a \langle \epsilon_J, \mathbb{Y}_t \rangle \mid \alpha_J^a = 0 \text{ if } J \cap (\mathcal{A} \setminus \{0, s, a, b\}) \neq \emptyset \right\}, \\ \mathcal{D}^{\text{sig},b} &:= \left\{ \delta_t^b = \langle \alpha^b, \mathbb{Y}_t \rangle := \sum_{|J| \leq K} \alpha_J^b \langle \epsilon_J, \mathbb{Y}_t \rangle \mid \alpha_J^b = 0 \text{ if } J \cap (\mathcal{A} \setminus \{0, s, a, b\}) \neq \emptyset \right\}.\end{aligned}$$

Now we show that restricting to Lin-Sig-Offsets allows us to recast the optimization problem (2.5) as a tractable optimization problem.

First, note that as a direct consequence of Proposition 2.6, for any Lin-Sig-Offset $\delta \in \mathcal{D}^{\text{sig}}$, both Z^δ and I are linear functionals of the signature of Y .

Corollary 2.8. For any $\delta = (\langle \alpha^a, \mathbb{Y} \rangle, \langle \alpha^b, \mathbb{Y} \rangle) \in \mathcal{D}^{\text{sig}}$,

$$Z_t^\delta = \langle (\alpha^a; \epsilon_a)^\sim + (\alpha^b; \epsilon_b)^\sim, \mathbb{Y}_t \rangle, \quad I_t = \langle (\epsilon_b - \epsilon_a; \epsilon_s)^\sim, \mathbb{Y}_t \rangle.$$

Remark 2.9. It is important to notice that two sources of dependence on δ are present. The first is explicit: the coefficients (α^a, α^b) appear directly in Z^δ . The second is implicit: through the intensity condition (2.4), the control also governs the law of Y and hence the distribution of its signature $\mathbb{Y}$, which enters both Z^δ and I .

To conclude, if U is polynomial, or well approximated by one, Corollary 2.8 allows us to recast (2.5) as

$$\sup_{\alpha^a, \alpha^b} \mathbb{E} \left[\tilde{U} \left(\langle (\alpha^a; \epsilon_a)^\sim + (\alpha^b; \epsilon_b)^\sim, \mathbb{Y}_T \rangle, \langle (\epsilon_b - \epsilon_a; \epsilon_s)^\sim, \mathbb{Y}_T \rangle \right) \right], \quad (2.6)$$

where $\tilde{U}(\zeta, \iota) = U(\zeta, \iota, \iota^2)$. By the shuffle property of the signature (2.3), the objective is then polynomial in (α^a, α^b) , with all dependence on the control law encoded in the expected signature $\mathbb{E}[\mathbb{Y}_T]$.

3 Mean-variance objective

In this section, we specialize the general framework to a mean-variance objective and study the resulting optimization problem via signature methods. This setting will serve as the basis for the numerical experiments in Section 4.

For any strategy δ , the objective function takes the form

$$J(\delta) := \mathbb{E} \left[Z_T^\delta \right] - \eta \mathbb{E} \left[(I_T)^2 \right], \quad \eta > 0, \quad (3.1)$$

where, recall that

$$dZ_t^\delta = \delta_t^a dN_t^a + \delta_t^b dN_t^b, \quad dI_t = Q_t dS_t, \quad Q_t = N_t^b - N_t^a.$$

Remark 3.1. When S is a $\mathbb{P}$ -martingale, (3.1) captures the classical trade-off between expected execution revenue and inventory risk.

3.1 Problem reformulation via signature methods

Restricting to Lin-Sig-Offsets $\delta \in \mathcal{D}^{\text{sig}}$, the optimization reduces to a search over the parameter $\alpha := (\alpha^a, \alpha^b)$. By Corollary 2.8,

$$Z_T^\delta = \left\langle (\alpha^a; \epsilon_a)^\sim + (\alpha^b; \epsilon_b)^\sim, \mathbb{Y}_T \right\rangle, \quad I_T = \left\langle (\epsilon_b - \epsilon_a; \epsilon_s)^\sim, \mathbb{Y}_T \right\rangle.$$

To lighten notation, we introduce the tensor-valued functionals

$$\ell_Z(\alpha) := (\alpha^a; \epsilon_a)^\sim + (\alpha^b; \epsilon_b)^\sim, \quad \ell_I := (\epsilon_b - \epsilon_a; \epsilon_s)^\sim,$$

so that $Z_T^\delta = \langle \ell_Z(\alpha), \mathbb{Y}_T \rangle$ and $I_T = \langle \ell_I, \mathbb{Y}_T \rangle$. Applying the shuffle identity (2.3) to the quadratic term in (3.1) gives

$$(I_T)^2 = \left\langle \ell_I \sqcup \ell_I, \mathbb{Y}_T \right\rangle,$$

and the objective (3.1) becomes

$$J(\alpha) = \left\langle L(\alpha), \mathbb{E}_\alpha[\mathbb{Y}_T] \right\rangle, \quad L(\alpha) := \ell_Z(\alpha) - \eta(\ell_I \sqcup \ell_I). \quad (3.2)$$

Pseudo-linear structure of the objective. Equation (3.2) reveals a *pseudo-linear structure* that is, to the best of our knowledge, new in the signature literature. As discussed in Remark 2.9, the dependence of $J(\alpha)$ on α operates through two distinct channels:

- (i) The functional $L(\alpha)$, which is explicit and polynomial in α , hence straightforward to differentiate.
- (ii) The expected signature $\mathbb{E}_\alpha[\mathbb{Y}_T]$, which depends (implicitly) on α through the controlled law of (N^a, N^b) , since the Lin-Sig-Offsets govern the arrival intensities via (2.4) and thus the entire trajectory distribution.

Crucially, this implicit dependence, does not obstruct gradient-based optimization: the gradient of $\mathbb{E}_\alpha[\mathbb{Y}_T]$ with respect to α is precisely what the REINFORCE estimator provides, as detailed in Algorithm 1 below.

3.2 Likelihood-ratio gradient

The goal is to find the parameter α , and hence the optimal offsets, that maximizes (3.2). We rewrite the objective as

$$J(\alpha) = \mathbb{E}_\alpha[R(\omega; \alpha)], \quad (3.3)$$

where

$$R(\omega; \alpha) := Z_T^\delta(\alpha) - \eta I_T^2 = \langle \ell_Z(\alpha), \mathbb{Y}_T(\omega) \rangle - \eta \langle \ell_I \sqcup \ell_I, \mathbb{Y}_T(\omega) \rangle \quad (3.4)$$

is the pathwise reward. As highlighted in Section 3.1, J depends on α through two distinct channels: directly, via the reward $R(\omega; \alpha)$, and implicitly, via the controlled law $\mathbb{P}_\alpha$, which redistributes probability mass across trajectories as the offsets vary. The latter dependence cannot be handled by a pathwise derivative, since the parameter shifts the underlying measure rather than a single realization; this is what motivates the likelihood-ratio approach adopted below.

Gradient formula. Suppose the family $(\mathbb{P}_\alpha)_\alpha$ is dominated by a common reference measure $\mathbb{P}^\circ$ on path space, with Radon–Nikodym density $p_\alpha(\omega) := d\mathbb{P}_\alpha/d\mathbb{P}^\circ$ that is strictly positive and differentiable in α for $\mathbb{P}^\circ$ -a.e. ω , and that both $R(\omega; \alpha)p_\alpha(\omega)$ and $R(\omega; \alpha)\nabla_\alpha p_\alpha(\omega)$ admit a locally dominated $\mathbb{P}^\circ$ -integrable envelope. These are standard conditions in the policy-gradient literature. Differentiating under the integral sign and applying the score-function identity $\nabla_\alpha p_\alpha = p_\alpha \nabla_\alpha \log p_\alpha$ yields

$$\nabla_\alpha J(\alpha) = \underbrace{\mathbb{E}_\alpha[\nabla_\alpha R(\omega; \alpha)]}_{\text{explicit algebraic term}} + \underbrace{\mathbb{E}_\alpha[R(\omega; \alpha) \nabla_\alpha \log p_\alpha(\omega)]}_{\text{likelihood-ratio term}}. \quad (3.5)$$

The first term arises from the direct dependence of the execution revenue on α through the tilde representation, and admits the closed-form expression

$$\nabla_\alpha R(\omega; \alpha) = \langle \nabla_\alpha \ell_Z(\alpha), \mathbb{Y}_T(\omega) \rangle.$$

The second accounts for the dependence of the controlled law $\mathbb{P}_\alpha$ on α , and is estimated via the REINFORCE estimator detailed in the following section.

3.3 Pseudo-algorithm

We summarize a generic gradient-ascent scheme for maximizing (3.3) on the parameters $\alpha = (\alpha^a, \alpha^b)$ following the ideas developed in Section 3.2.

Input: Parameters: Truncation level K ; batch size B ; learning-rate schedule $(\rho_m)_{m \geq 0}$; penalty weight $\eta > 0$; maximum iterations $M_{\max}$; gradient-clip $G_{\max}$; tolerance $\varepsilon > 0$.

Initialisation: $\alpha^{(0)} = (\alpha^{a,(0)}, \alpha^{b,(0)}) \in (T^K(\mathbb{R}^d))^2$ (e.g. $\alpha^{(0)} = \mathbf{0}$); baseline $\bar{R}^{(0)} = 0$; counter $m \leftarrow 0$.

repeat

/* 1. Simulate trajectories under the current policy. */

Draw B independent paths $\omega^{(1)}, \dots, \omega^{(B)} \sim \mathbb{P}_{\alpha^{(m)}}$;

Compute $\mathbb{Y}_T^{(b)}$ and prefix features $(\phi_n^{(b)})_{n=0}^{N-1}$ for each $b = 1, \dots, B$;

/* 2. Evaluate the pathwise reward. */

For $b = 1, \dots, B$, set

$$R^{(b)} := \langle \ell_Z(\alpha^{(m)}), \mathbb{Y}_T^{(b)} \rangle - \eta \langle \ell_I \sqcup \ell_I, \mathbb{Y}_T^{(b)} \rangle, \quad \ell_I := (\epsilon_b - \epsilon_a; \epsilon_s)^\sim;$$

/* 3. Compute the algebraic gradient of the reward. */

For $b = 1, \dots, B$, set

$$g_{\text{alg}}^{(b)} := \nabla_{\alpha} R^{(b)} = \langle \nabla_{\alpha} \ell_Z(\alpha^{(m)}), \mathbb{Y}_T^{(b)} \rangle;$$

/* 4. Update the variance-reduction baseline. */

$$\bar{R}^{(m)} \leftarrow (1 - \beta) \bar{R}^{(m-1)} + \beta \frac{1}{B} \sum_{b=1}^B R^{(b)}, \quad \beta \in (0, 1];$$

Set $\tilde{R}^{(b)} := R^{(b)} - \bar{R}^{(m)}$ for each b ;

/* 5. Evaluate the score function. */

For $b = 1, \dots, B$, compute $\nabla_{\alpha} \log p_{\alpha^{(m)}}(\omega^{(b)})$ via (4.4);

/* 6. Assemble the policy-gradient estimator. */

$$\widehat{\nabla_{\alpha} J}^{(m)} = \frac{1}{B} \sum_{b=1}^B \left(g_{\text{alg}}^{(b)} + \tilde{R}^{(b)} \nabla_{\alpha} \log p_{\alpha^{(m)}}(\omega^{(b)}) \right);$$

/* 7. Clipped gradient ascent. */

$$g \leftarrow \min(1, G_{\max} / \|\widehat{\nabla_{\alpha} J}^{(m)}\|) \widehat{\nabla_{\alpha} J}^{(m)};$$

$$\alpha^{(m+1)} \leftarrow \alpha^{(m)} + \rho_m g;$$

$$m \leftarrow m + 1;$$

until $\|\widehat{\nabla_{\alpha} J}^{(m)}\| < \varepsilon$ or $m = M_{\max}$;

Return: Optimal parameters $\alpha^* = \alpha^{(m)}$.

Algorithm 1: Sig-REINFORCE algorithm

Remark 3.2. Notice that the above optimization principle applies to all intensity models and price dynamics for which the score of the controlled law is available.

4 Numerical experiments

We finally turn to the implementation and testing of the Sig-REINFORCE algorithm described in Section 3.3. We simulate a batch (state process) under the current policy, evaluate the pathwise reward, and combine an explicit algebraic gradient with a likelihood-ratio score. In this section we generate path of the price process in a Bachelier model, $S_t = S_0 + \sigma W_t$, with

$S_0, \sigma > 0$.

4.1 Experimental design and the simulation engine

Two test regimes. We evaluate the method in two order flows regimes. In the first, order flow is conditionally Poisson with spread-controlled intensities and the price follows a Bachelier model; this admits a closed-form mean–variance optimum [Cartea et al. \(2015\)](#); [Guéant et al. \(2012\)](#), which we use as an *expert* benchmark to check that Sig-REINFORCE recovers a known optimum (Section 4.3). In the second, order flow is a self-exciting Hawkes process (Section 4.4): genuinely path-dependent and without a closed-form solution, yet still tractable, so that it stresses the path-dependent features of the signature policy. In both regimes we benchmark against a PPO baseline (Section 4.2). Note that the linearization of Section 2.4 is not tied to this choice of processes since the only requirement for Sig-REINFORCE is a counting process whose intensity is positive and differentiable in the quoted spreads, so that the score $\nabla_{\alpha} \log p_{\alpha}$ exists; any such model is admissible by swapping the score alone (Remark 4.3).

Dropping the price and quadratic-variation channels. The augmented state Y of Section 2.4 carries the price S and its quadratic variation $[S]$ alongside the counting channels. In the experiments we set to zero all policy coefficients on words containing the letters s and qs , restricting the policy alphabet to $\{0, a, b\}$. This is not a limitation of the method but a variance-control and computational choice tailored to the benchmark. Under the Bachelier dynamics $S_t = S_0 + \sigma W_t$ the price is a martingale, independent of the order flow, and the mean-variance optimum is a function of the time-to-maturity and the inventory $Q = N^b - N^a$ only, both recoverable from the reduced signature on $\{0, a, b\}$. The omitted coefficients carry no true signal for the optimal spread against this benchmark and would only enlarge the parameter vector and inflate the gradient variance. In the reduced setting the price still enters the objective through the inventory mark-to-market, as described below.

Generative data and streaming signature. Each path carries the increment vector $dX_t = (dt, dN_t^a, dN_t^b)$ on the alphabet $\{0, a, b\}$, with unit jumps; the stock price is carried over when needed to compute the pathwise reward for output processing. The truncated signature $\mathbb{Y}_t = \pi_{\leq K}(\cdot)$ is updated incrementally at every step, and the quoted half-spreads are read off as the linear functionals $\delta_t^a = \langle \alpha^a, \mathbb{Y}_t \rangle$ and $\delta_t^b = \langle \alpha^b, \mathbb{Y}_t \rangle$ of Definition 2.7. We fix the truncation level $K = 3$ throughout, a choice we justify below.

Choice of truncation level. A signature-linear policy is indexed by the words of length at most K on $\{0, a, b\}$, whose number grows geometrically as $\sum_{j=0}^K 3^j$ per side, so each extra level enlarges the policy class but also inflates the dimension, hence practically the variance, of the likelihood-ratio gradient. We choose the smallest level that is already expressive enough by a representability test independent of training: we generate paths under the closed-form expert [Cartea et al. \(2015\)](#), compute the truncated signature of the augmented path, and ridge-regress the expert’s quoted half-spread on the signature features, reporting the in-sample R^2 of the fit and the out-of-sample percentage mean–variance reward of the replayed linear-in-signature rule (Figure 1). The gain from $K = 2$ to $K = 3$ is substantial and $K = 3$ already recovers essentially the full expert reward, whereas $K = 4$ adds only marginal value at a much larger parameter count. We therefore fix $K = 3$.

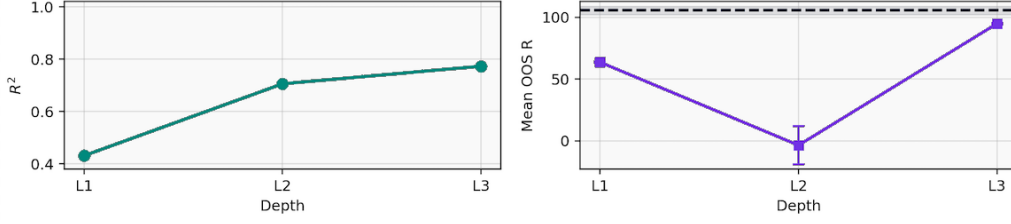


Figure 1: R^2 of the ridge fit to the expert policy and out-of-sample percentage reward of the fitted linear-in-signature rule, as a function of the signature truncation level K , for the Poisson order flow.

Conditional independence of the state processes. The Brownian motion is pre-sampled once per batch on a fine grid and then frozen: the price path is the deterministic functional $S_t = S_0 + \sigma W_t$ obtained by interpolation, and the same W is reused across all methods and the expert. This common-random-numbers device removes price noise from every method comparison. The order flow is drawn from an independent random source, so that, conditionally on $\mathcal{F}_T^W$ and the predictable intensities, N^a and N^b share no driving noise and are therefore conditionally independent by Watanabe’s characterisation (Brémaud, 1981, Ch. II, Thm. T7). The engine realises this structure exactly: the next arrival is obtained by superposition, a single waiting time drawn from $\lambda^a + \lambda^b$, with the side assigned with probability $\lambda^a/(\lambda^a + \lambda^b)$. The two sides remain coupled only through the predictable intensities, each of which sees the other’s past through the signature, but never through their innovations. This is precisely the structure that makes the joint likelihood factor across sides, as in (4.1), and the score split into the decoupled parameter blocks α^a, α^b .

4.2 Sig-REINFORCE v.s. Sig-PPO

In this section we contrast the two learning procedures used in Sections 4.3–4.4. Both Sig-PPO and Sig-REINFORCE share the same environment, the same simulator and the same signature feature map. They differ only in how the gradient of the expected reward $J(\alpha) = \mathbb{E}[R_T(\alpha)]$ is estimated from rollouts. We deliberately pick PPO as the deep-RL baseline because it is an *on-policy* method: each gradient update is computed from rollouts generated by (a recent copy of) the current policy, so no replay buffer or behaviour-policy correction is involved. This is the same regime in which the likelihood-ratio identity (4.3) is unbiased, and it makes the comparison genuinely method-vs-method. Both estimators are built from the same score function $\nabla_\alpha \log p_\alpha$; the differences are how the return is attributed, accumulated over the whole path or through a combination of step reward and critic.

Sig-REINFORCE (path-wise score). We estimate $\nabla_\alpha J(\alpha)$ through the explicit part and a single likelihood-ratio applied to the whole trajectory of W, N^a, N^b :

$$\mathbb{E} \left[(R(\alpha) - R^\circ) \nabla_\alpha \log p_\alpha(W, N^a(\alpha), N^b(\alpha)) \right] = \mathbb{E}[E_\alpha(W)],$$

where for any $\varsigma \in \mathcal{C}([0, T]; \mathbb{R})$ (a path on the Wiener space)

$$E_\alpha(\varsigma) := \mathbb{E} \left[(R_T(\alpha) - R^\circ) \nabla_\alpha \log p_\alpha(\varsigma, N^a(\alpha), N^b(\alpha)) \right]^1,$$

where R° is a path-independent baseline to compute rewards (sample mean across the rollout batch) and the score is exactly (4.3). The gradient is collected once per trajectory after the

¹Note that the dependence with respect to W is also hidden in the reward and α through $\omega \in \Omega$.

terminal reward is observed, and each batch produces one gradient step. Every component of the estimator is closed-form in α : no value function is learned and no auxiliary network is introduced.

Sig-PPO (step-wise advantage). PPO, see [Schulman et al. \(2017\)](#), maximizes the same objective J , but factorises it through the per-step return decomposition $R_T = \sum_n r_{n+1}$ and replaces the path-wise score by per-step advantage estimates.

A learned critic $V_\theta(\mathbb{Y}_{\tau_n})$ approximates the state value, the advantage

$$\hat{A}_n = \sum_{m \geq 0} (\gamma \lambda_{\text{GAE}})^m [r_{n+m+1} + \gamma V_\theta(\mathbb{Y}_{\tau_{n+m+1}}) - V_\theta(\mathbb{Y}_{\tau_{n+m}})]$$

is computed via generalised advantage estimation, and the actor is updated by gradient ascent on the clipped surrogate

$$L^{\text{clip}}(\alpha) = \mathbb{E} \left[\min(\rho_n \hat{A}_n, \text{clip}(\rho_n, 1-\epsilon, 1+\epsilon) \hat{A}_n) \right], \quad \rho_n = \frac{p_\alpha(\text{event at } \tau_{n+1} | \mathcal{F}_{\tau_n})}{p_{\alpha_{\text{old}}}(\text{event at } \tau_{n+1} | \mathcal{F}_{\tau_n})}.$$

Each rollout produces M per-step gradients (one per event), reused across several inner optimization epochs, with the clipping ensuring monotone improvement under bounded policy drift.

Key contrasts REINFORCE vs PPO.

- *Granularity.* Sig-REINFORCE assigns the full terminal advantage $R_T - b$ to every $\nabla_\alpha \log p$ contribution from each side and event; Sig-PPO assigns a step-local advantage $\hat{A}_n$ to each $\nabla_\alpha \log p(\text{event}_n)$. The Sig-PPO estimator is *lower variance* per gradient step when the critic is accurate, at the cost of bias from value-function approximation. Our estimator is unbiased by construction but has higher variance on long horizons, which is why we use the pathwise control variates.
- *Use of the structure.* The score (4.3) exploits the existence of an underlying (differentiable) functional relationship between spread and intensity which in this case is in closed form, yielding a clean score form). Sig-PPO is model-agnostic and treats the policy as a black-box density; it does not see the model and so cannot enjoy the same variance reduction from algebraic cancellation.
- *Computational complexity.* Sig-REINFORCE performs one outer update per batch of trajectories; Sig-PPO performs several minibatch updates of the same data per outer iteration. For matched env-step budgets, Sig-PPO brings the computational cost an order of magnitude up with respect to Sig-REINFORCE.

4.3 Poisson order flow

As a sanity check of our method, we aim at comparing this optimization problem with an explicit benchmark setting for a choice of price model S and intensity $\delta \mapsto \lambda(\delta)$ linking the quoted spreads to the order-arrival intensities. We adopt here the classical setting of [Avellaneda and Stoikov \(2008\)](#); [Guéant et al. \(2012\)](#), under which the optimal market-making policy admits a known closed-form solution, see [Cartea et al. \(2015\)](#). This serves as a controlled benchmark: it lets us test whether our algorithm recovers the optimum in a regime where the analytical solution is available. We stress, however, that the algorithm itself is not tied to this specification, the algebraic signature linearization below is purely structural, and the score (4.2)–(4.4) can be replaced accordingly under more general dynamics (see Remark 4.3 below). Therefore,

we assume that N^a and N^b are conditionally Poisson processes, the arrival intensities depend exponentially on the quoted spreads: $\lambda_t^a(\delta_t^a) = Ae^{-k\delta_t^a}$ and $\lambda_t^b(\delta_t^b) = Ae^{-k\delta_t^b}$, where $A, k > 0$. Fix the filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ with $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ the natural augmented filtration of (W, N^a, N^b) . By construction the signature $\mathbb{Y}_t$ is $\mathcal{F}_t$ -measurable, so the quoted half-spreads $\delta_t^{a/b}(\alpha) = \langle \alpha^{a/b}, \mathbb{Y}_{t-} \rangle$ are $\mathbb{F}$ -predictable², and hence so are the controlled intensities

$$\lambda_t^a(\alpha) = Ae^{-k\delta_t^a(\alpha)}, \quad \lambda_t^b(\alpha) = Ae^{-k\delta_t^b(\alpha)}.$$

The Doob–Meyer decomposition gives the $\mathbb{F}$ -compensators $\Lambda_t^s(\alpha) = \int_0^t \lambda_r^s(\alpha) dr$, so that

$$M_t^s(\alpha) := N_t^s - \Lambda_t^s(\alpha), \quad s \in \{a, b\}.$$

are zero-mean $\mathbb{F}$ -local martingales. Under our specification W, N^a, N^b have no shared driving noise, so by Watanabe’s characterisation theorem (Brémaud, 1981, Ch. II, Thm. T7) N^a and N^b are conditionally independent given $\mathcal{F}_T^W$ (the Brownian sigma-algebra) and the predictable intensities.

Joint likelihood under α and score function. Let $\mathbb{P}^0$ denote the reference measure under which both N^a and N^b are unit-rate Poisson and independent of W . The Girsanov/Watanabe theorem for marked point processes (Daley and Vere-Jones, 2003, Thm. 7.4.I) gives the Radon–Nikodym derivative

$$\left. \frac{d\mathbb{P}_\alpha}{d\mathbb{P}^0} \right|_{\mathcal{F}_T} = \prod_{s \in \{a, b\}} \exp \left\{ \int_0^T \log \lambda_t^s(\alpha) dN_t^s - \int_0^T (\lambda_t^s(\alpha) - 1) dt \right\}.$$

Conditioning on $\mathcal{F}_T^W$, the product over $s \in \{a, b\}$ collapses to a sum in log-space so that for any $\varsigma \in \mathbb{R}$

$$\begin{aligned} \log p_\alpha(\varsigma, N^a, N^b | \mathcal{F}_T^W) &= \int_0^T \log \lambda_t^a(\alpha) dN_t^a - \int_0^T \lambda_t^a(\alpha) dt \\ &\quad + \int_0^T \log \lambda_t^b(\alpha) dN_t^b - \int_0^T \lambda_t^b(\alpha) dt + \text{const}(\alpha), \end{aligned} \quad (4.1)$$

and remembering that α is a function of ς (the frozen realization of a path of W in the conditioning), the paths N^a, N^b . Under standard integrability of the intensities we exchange ∇_α with the stochastic integral against dN and the Lebesgue integral. For each side, since $\log \lambda_t^{a/b} = \log A - k\delta_t^{a/b}$,

$$\nabla_\alpha \log \lambda_t^{a/b}(\alpha) = -k \nabla_\alpha \delta_t^{a/b}(\alpha), \quad \nabla_\alpha \lambda_t^{a/b}(\alpha) = -k \lambda_t^{a/b}(\alpha) \nabla_\alpha \delta_t^{a/b}(\alpha).$$

Substituting into the derivative of (4.1) and grouping by side, the dependence on $\nabla_\alpha \delta_t^{a/b}$ assembles into the compensated counting differential $dM_t^{a/b} = dN_t^{a/b} - \lambda_t^{a/b} dt$:

$$\nabla_\alpha \log p_\alpha = -k \int_0^T \nabla_\alpha \delta_t^a(\alpha) (dN_t^a - \lambda_t^a(\alpha) dt) - k \int_0^T \nabla_\alpha \delta_t^b(\alpha) (dN_t^b - \lambda_t^b(\alpha) dt). \quad (4.2)$$

Because $\delta_t^{a/b} = \langle \alpha^{a/b}, \mathbb{Y}_t \rangle$ is linear, the gradient collapses to the signature features themselves:

$$\nabla_{\alpha^a} \delta_t^a = \mathbb{Y}_t, \quad \nabla_{\alpha^b} \delta_t^b = \mathbb{Y}_t, \quad \nabla_{\alpha^a} \delta_t^b = 0, \quad \nabla_{\alpha^b} \delta_t^a = 0,$$

²Continuous in t between events, so left and right limits agree where the signature is continuous; the values just before each jump time are $\mathcal{F}_{t-}$ -measurable, which is what predictability of the intensity requires. See (Brémaud, 1981, Ch. II S2).

and (4.2) factorises across the two parameter blocks:

$$\nabla_{\alpha^a} \log p_{\alpha} = -k \int_0^T \mathbb{Y}_t (dN_t^a - \lambda_t^a dt), \quad \nabla_{\alpha^b} \log p_{\alpha} = -k \int_0^T \mathbb{Y}_t (dN_t^b - \lambda_t^b dt). \quad (4.3)$$

Both integrals are $\mathbb{F}$ -martingale transforms of the compensated counting processes $M^{a/b}$, so the score has mean zero under $\mathbb{P}_{\alpha}$ — the standard identity $\mathbb{E}[\nabla \log p_{\alpha}] = 0$, which is what makes the Sig-REINFORCE estimator unbiased.

Remark 4.1. On a grid $t_n = n dt$, with

$$\Delta N_{n+1}^{a/b} := N_{t_{n+1}}^{a/b} - N_{t_n}^{a/b},$$

and prefix features $\phi_n = \pi_{\leq K}(\mathbb{Y}_{t_n})$, or their flattened representation, (4.3) becomes

$$\nabla_{\alpha^a} \log p_{\alpha} = -k \sum_{n=0}^{N-1} \phi_n (\Delta N_{n+1}^a - \lambda_n^a dt), \quad \nabla_{\alpha^b} \log p_{\alpha} = -k \sum_{n=0}^{N-1} \phi_n (\Delta N_{n+1}^b - \lambda_n^b dt), \quad (4.4)$$

where $\lambda_n^{a/b} := \lambda_{t_n}^{a/b}(\delta_{t_n}^{a/b})$.

Remark 4.2. Indexed by the event sequence $0 = \tau_0 < \tau_1 < \dots < \tau_M = T$ of ordered ask and bid arrival times (with $\tau_M := T$ as a terminal sentinel), with inter-event intervals

$$\Delta \tau_n := \tau_{n+1} - \tau_n, \quad \Delta N_{n+1}^{a/b} \in \{0, 1\}$$

the indicator that the event at τ_{n+1} is of type a or b , and event-time features $\phi_n = \pi_{\leq K}(\mathbb{Y}_{\tau_n})$, (4.3) becomes

$$\nabla_{\alpha^a} \log p_{\alpha} = -k \sum_{n=0}^{M-1} \phi_n (\Delta N_{n+1}^a - \lambda_n^a \Delta \tau_n), \quad \nabla_{\alpha^b} \log p_{\alpha} = -k \sum_{n=0}^{M-1} \phi_n (\Delta N_{n+1}^b - \lambda_n^b \Delta \tau_n),$$

where $\lambda_n^{a/b} := \lambda^{a/b}(\delta_{\tau_n}^{a/b})$. Under our assumption, λ^a and λ^b are piecewise constant on each random inter-event interval $[\tau_n, \tau_{n+1})$, so $\lambda_n^{a/b} \Delta \tau_n$ is the exact compensator increment $\int_{\tau_n}^{\tau_{n+1}} \lambda_s^{a/b} ds$.

We now compare numerically in this setting the Sig-REINFORCE algorithm with the Sig-PPO method described in Section 4.2, and both of them against the analytical optimal solution of Cartea et al. (2015). We instantiate the benchmark with the following values: horizon $T = 300$ s, spread-coupling $k = 0.2$, baseline order-arrival rate $A = 0.517 \text{ s}^{-1}$, price volatility $\sigma = 2$ (in dollars per $\sqrt{\text{s}}$) so that the Bachelier mid is $S_t = S_0 + \sigma W_t$ with $S_0 = 100$ and a tick size of 0.01. The risk-aversion parameter of the mean-variance reward is $\gamma = 0.05$ and the per-side inventory cap is $Q_{\max} = 10$. Figure 2 compares, on the left, the average executed orders (total order flow) of the market maker and, on the right, their optimal quotes, obtained using Sig-REINFORCE, Sig-PPO, and the explicit analytical solution (the expert). We first observe a very similar pattern between Sig-REINFORCE and Sig-PPO relative to the benchmark. While the expert remains fairly stable, both Sig-REINFORCE and Sig-PPO exhibit more variance in the best ask/bid quotes. Notably, Sig-REINFORCE achieves this with a slightly lower quote variance than Sig-PPO. The bid/ask spreads are also wider under Sig-REINFORCE and Sig-PPO, consequently leading to a lower average total order flow.

We now turn to the reward (payoff) of the mean-variance problem. We simulate it over several fresh trajectories at the optimal values in Figure 3 for Sig-REINFORCE, Sig-PPO, and the expert, and we also report the value of the problem at time 300 in the box at the top right. While the reward obtained by the expert is, as expected, the highest, the gap with Sig-REINFORCE and Sig-PPO remains small, with Sig-REINFORCE achieving a higher mean reward than Sig-PPO (+425.29 versus +417.73).

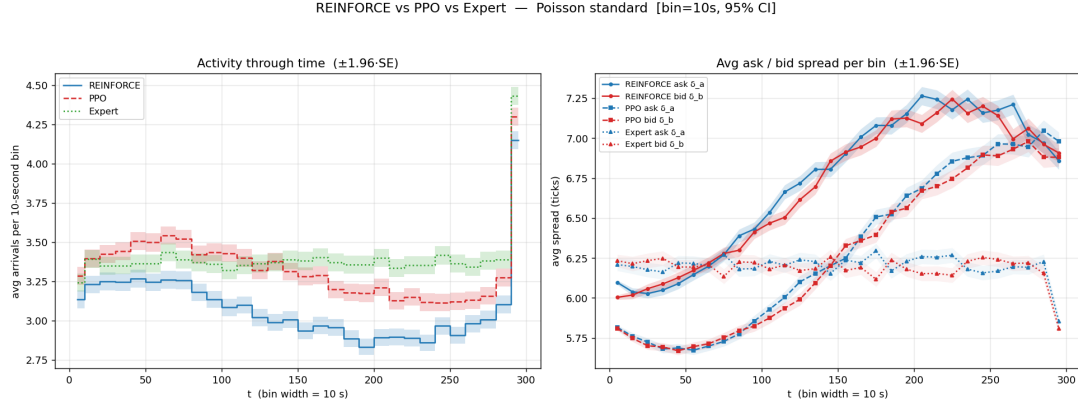


Figure 2: Averaged market order arrivals (left) and average bid and ask spread quoted (right) with Sig-REINFORCE, Sig-PPO and Expert (analytical solutions) for Poisson order flows.

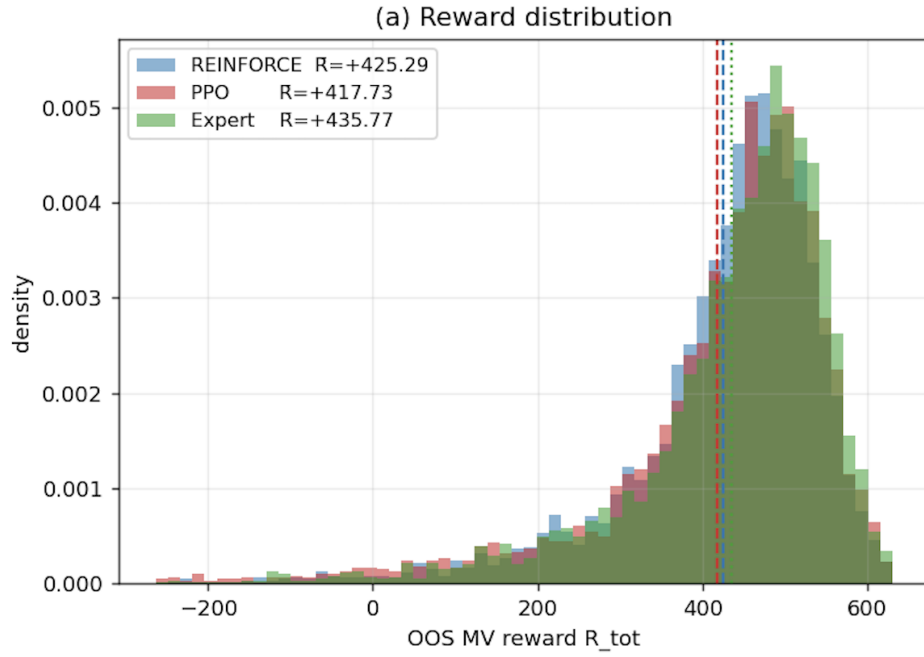


Figure 3: Payoff distribution over time of the market maker with Sig-REINFORCE, Sig-PPO and Expert (analytical solutions) for Poisson order flows.

4.4 Hawkes order flow

We now turn to the case where the order flow follows a Hawkes process. Preliminary works on market making with self-exciting order flows have been investigated in [Jusselin \(2021\)](#), but the theoretical study of the mean-variance problem remains open. This section focuses on the numerical analysis of this problem with Sig-REINFORCE algorithm compared with Sig-PPO. Let N^a, N^b be two counting process with self-excited intensity given by

$$\lambda^a(t, \delta^a) = e^{-k\delta^a}(\mu_a + X_t^a), \quad \lambda^b(t, \delta^a) = e^{-k\delta^b}(\mu_b + X_t^b),$$

where

$$dX_t^a = -\beta X_t^a dt + dN_t^a, \quad dX_t^b = -\beta X_t^b dt + dN_t^b, \quad X_0^a = X_0^b = 0.$$

or equivalently

$$X_t^a = \sum_{t_i^a < t} e^{-\beta(t-t_i^a)}, \quad X_t^b = \sum_{t_i^b < t} e^{-\beta(t-t_i^b)},$$

where t_i^a, t_i^b denotes the i th event time of N^a, N^b respectively. In the numerical experiments we choose $T = 300$, $k = 0.2$ and $\mu_a = \mu_b = 0.3102$, $\beta = 2$, so that $n = 1/\beta = 0.5$, and the intensity reduces to

$$\lambda^s(t, \delta^s) = e^{-0.2\delta^s} \left(0.3102 + \sum_{t_i^s < t} e^{-2(t-t_i^s)} \right), \quad t \in [0, 300], \quad s \in \{a, b\}.$$

Remark 4.3 (Score for the Hawkes scenario). *The Hawkes process is still a point process with predictable intensity, so the Watanabe likelihood (4.1) applies verbatim with $\lambda_t^s(\alpha) = e^{-k\delta_t^s(\alpha)}(\mu_s + X_t^s)$. The only conceptual change is that λ_t^s now depends on the endogenous memory X_t^s , which is itself a deterministic functional of the observed history $(N_u^a, N_u^b)_{u < t}$. In particular X^s is $\mathbb{F}$ -predictable, so λ^s is too, and the compensator $\Lambda_t^s(\alpha) = \int_0^t \lambda_u^s(\alpha) du$ is well-defined sample-path by sample-path.*

Per-step score. Index by the event sequence $0 = t_0 < t_1 < \dots < t_M = T$ as in (4.4), with inter-event interval $\Delta t_n := t_{n+1} - t_n$, increments $\Delta N_n^{a/b} \in \{0, 1\}$, and truncated event-time features $\phi_n := \pi_{\leq K}(\mathbf{Y}_{t_n})$, so that the quote $\delta_{t_n}^s = \langle \alpha^s, \phi_n \rangle$ and the memory $x_s := X_{t_n}^s$ are both $\mathcal{F}_{t_n}$ -measurable. The memory decays deterministically between events, $X_u^s = x_s e^{-\beta(u-t_n)}$ for $u \in [t_n, t_{n+1})$, so the exact compensator increment integrates in closed form:

$$\Delta \Lambda_n^s = \int_{t_n}^{t_{n+1}} \lambda_u^s(\alpha) du = e^{-k\delta_{t_n}^s(\alpha)} \left[\mu_s \Delta t_n + x_s \frac{1 - e^{-\beta \Delta t_n}}{\beta} \right].$$

Conditioning on $\mathcal{F}_{t_n}$ freezes x_s and treats it as a fixed input, so λ_u^s depends on the parameters only through the multiplicative factor $e^{-k\delta_{t_n}^s(\alpha)}$. Differentiating the local log-likelihood contribution $\Delta N_n^s \log \lambda_{t_n}^s - \Delta \Lambda_n^s$ yields

$$\frac{\partial \log \mathcal{L}_s^{(n)}}{\partial \delta_{t_n}^s} = -k (\Delta N_n^s - \Delta \Lambda_n^s), \quad \nabla_{\alpha^s} \log \mathcal{L}_s^{(n)} = -k \phi_n (\Delta N_n^s - \Delta \Lambda_n^s),$$

and summing over the event grid recovers the truncated-feature analogue of (4.3),

$$\nabla_{\alpha^s} \log p_{\alpha} = -k \sum_{n=0}^{M-1} \phi_n (\Delta N_n^s - \Delta \Lambda_n^s), \quad s \in \{a, b\}.$$

Structurally identical to the Poisson score, with the only difference that $\Delta \Lambda_n^s$ now carries the memory term $x_s(1 - e^{-\beta \Delta t_n})/\beta$. Note that our score methodology is suitable for Hawkes processes since it is calibrated on the full path of the canonical process itself.

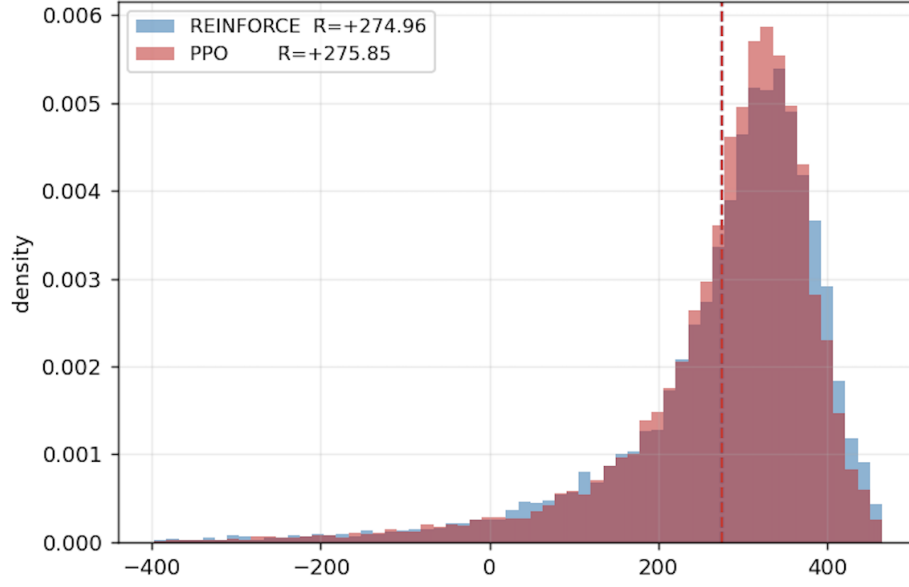


Figure 4: Return Sig-REINFORCE v.s. Sig-PPO

REINFORCE vs PPO — Hawkes standard [bin=10s, 95% CI]

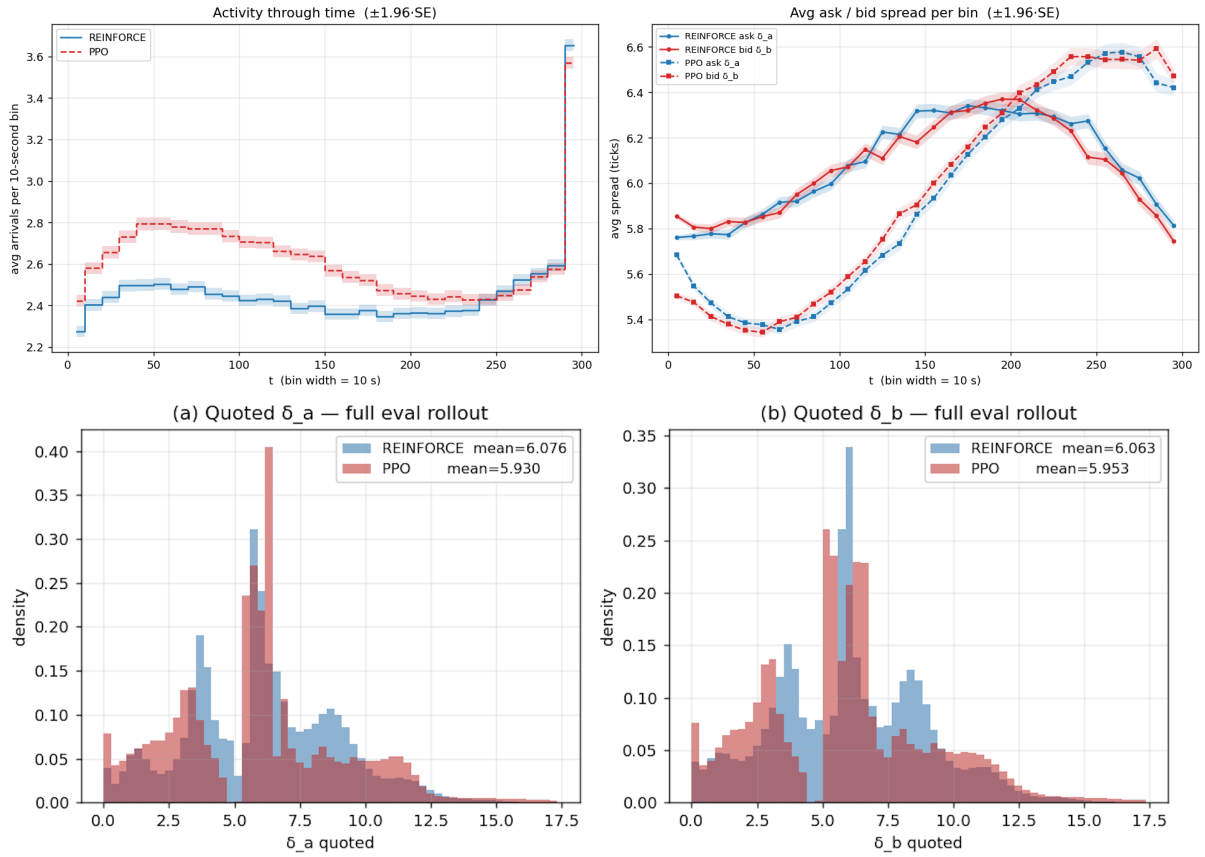


Figure 5: Expected Bid-Ask spread value with Sig-REINFORCE v.s. Sig-PPO (top) and respective distributions (bottom)

The reward distribution associated to the mean-variance problem are presented in Figure 4. The average reward with Sig-REINFORCE is statistically indistinguishable from Sig-PPO, which shows how it is possible to learn equally well looking at entire paths. Digging deeper into the policy found by the two algorithms, figure 5 compared the bid-ask spread obtained with our Sig-REINFORCE method and the one derived from Sig-PPO. We first observed that on a reasonable trading scale of $T = 300s$, Sig-REINFORCE has slightly less variance than Sig-PPO, especially close to the end of the horizon, similarly to the Poisson case previously investigated. This is confirmed by the distribution graphs which seems to be a bit more bimodal for Sig-PPO compared with Sig-REINFORCE. The average spread is again slightly higher with Sig-REINFORCE compared with Sig-PPO, but the shapes of the average quote in time for both algorithm is consistent.

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